

Exact Coordinates for Optimal Heilbronn Configurations

$n = 5, \dots, 9$ on the Unit Square

Companion to: N. Sudermann-Merx, *From Computational Certification to Exact Coordinates: Heilbronn*

This document lists the certified optimal coordinates for Heilbronn's triangle problem on the unit square $[0, 1]^2$ for $n = 5, \dots, 9$. All configurations are globally optimal, certified by solving a mixed-integer quadratically constrained program (MIQCP) with Gurobi and then deriving exact symbolic coordinates with SymPy. The derivation is fully automated by the script `exact_coordinates.py` included in this repository.

Points are labeled $1, \dots, n$. We denote the optimal minimum triangle area by Δ_n .

$$n = 5: \quad \Delta_5 = \frac{\sqrt{3}}{9} \approx 0.19245$$

Four critical triangles. Unique optimal configuration.

Pt	x	y
1	0	$\frac{1}{3}$
2	$\frac{\sqrt{3}}{3}$	0
3	1	$1 - \frac{\sqrt{3}}{3}$
4	$\frac{2}{3}$	1
5	0	1

$$n = 6: \quad \Delta_6 = \frac{1}{8} = 0.125$$

Six critical triangles. One-parameter family of optimal solutions parameterized by $c \in [0, \frac{1}{4}]$.

Pt	x	y
1	0	c
2	$\frac{1}{2}$	0
3	1	$\frac{1}{2} - c$
4	$\frac{1}{2}$	1
5	0	$c + \frac{1}{2}$
6	1	$1 - c$

$$n = 7: \quad \Delta_7 = -\frac{a}{2a-4} \approx 0.08386$$

Eight critical triangles. Coordinates are rational functions of $a \approx 0.2873$, the unique root in $(0, 1)$ of the irreducible cubic

$$a^3 + 5a^2 - 5a + 1 = 0.$$

Pt	x	y
1	0	0
2	$\frac{1}{a^2 - 3a + 2}$	0
3	1	a
4	$\frac{2a - 1}{a(a - 2)}$	1
5	0	1
6	$\frac{a - 1}{a - 2}$	a
7	$\frac{2(-a^2 + 3a - 1)}{a^3 - 4a^2 + 5a - 2}$	$-\frac{2a}{a - 1}$

$$n = 8: \quad \Delta_8 = -\frac{1}{36} + \frac{\sqrt{13}}{36} \approx 0.07238$$

Twelve critical triangles. Point-symmetric configuration with unique optimal solution involving $\sqrt{13}$.

Pt	x	y
1	0	0
2	$\frac{1}{6} + \frac{\sqrt{13}}{6}$	0
3	1	$\frac{7}{18} - \frac{\sqrt{13}}{18}$
4	1	1
5	0	$\frac{11}{18} + \frac{\sqrt{13}}{18}$
6	$\frac{5}{6} - \frac{\sqrt{13}}{6}$	1
7	$\frac{5}{6} - \frac{\sqrt{13}}{6}$	$\frac{7}{9} - \frac{\sqrt{13}}{9}$
8	$\frac{1}{6} + \frac{\sqrt{13}}{6}$	$\frac{2}{9} + \frac{\sqrt{13}}{9}$

$$n = 9: \quad \Delta_9 = -\frac{11}{64} + \frac{9\sqrt{65}}{320} \approx 0.05488$$

Eleven critical triangles. Configuration with four-fold rotational symmetry; unique optimal solution involving $\sqrt{65}$.

Pt	x	y
1	0	$1 - \frac{\sqrt{65}}{10}$
2	$\frac{3}{8} - \frac{\sqrt{65}}{40}$	0
3	1	$\frac{9}{16} - \frac{3\sqrt{65}}{80}$
4	$\frac{3}{8} - \frac{\sqrt{65}}{40}$	1
5	0	$\frac{5}{8} + \frac{\sqrt{65}}{40}$
6	$\frac{1}{4} + \frac{\sqrt{65}}{20}$	$\frac{3}{4} - \frac{\sqrt{65}}{20}$
7	$\frac{7}{16} + \frac{3\sqrt{65}}{80}$	0
8	$\frac{\sqrt{65}}{10}$	1
9	1	$\frac{5}{8} + \frac{\sqrt{65}}{40}$

Source: <https://github.com/spiralulam/heilbronn>